

Problem G. Edge Flipping

TimeLimit: 1 second
MemoryLimit: 1024 megabytes

You are given an undirected simple graph with n vertices and m edges. Each edge is labeled either 0 or 1. You may perform the following operation any number of times:

- Choose a vertex v .
- Flip the label of every edge incident to v . That is, every label 0 becomes 1, and every label 1 becomes 0.

Determine whether it is possible to make the labels of all edges equal to 0.

Input

The first line contains two integers n and m — the number of vertices and edges. Each of the next m lines contains three integers u , v , and c , indicating that there is an undirected edge between vertices u and v with label c .

The graph is simple: there are no self-loops or multiple edges between the same pair of vertices.

- $2 \leq n \leq 2 \times 10^5$
- $1 \leq m \leq \min\left(2 \times 10^5, \frac{n(n-1)}{2}\right)$
- $1 \leq u, v \leq n, u \neq v$
- $c \in \{0, 1\}$

Output

Print YES if it is possible to make every edge label equal to 0 using the operation any number of times. Otherwise, print NO.

Examples

standard input	standard output
4 4 1 2 1 2 3 0 3 4 1 1 4 0	YES
3 3 1 2 1 2 3 0 1 3 0	NO